

Controller & Driver



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Introduction

Integrated Controller & Driver, MMDC Series

MMDC-ST466 is a four-axis motion controller with built-in driver for two-phase bi-polar step motor operation, which is a technical know-how-intensive product.

By embedding multiple drivers in a single controller, the space savings and lower costs are realized due to its small size, and its usability is very convenient.



MMDC-ST466



MMDC-ST466-P



MMDC-SM466

► Feature

- **microstep driver with 51200 resolution (51200PPR)**

With microstep driver as standard, smoothness, low vibration, and high precision are achieved.

- **current conversion function of motor**

A simple S/W command entry allows you to set the current between 0.2A and 1.5A/Phase in 30 steps.

- **Various Communication Port**

The MMDC-ST466 controls the motor's behavior according to the command that is passed through the communication port on the High Control Unit (PC). Serial communication is available via RS-232 and USB, and Ethernet communication is also possible using LAN ports.

Also, 15 additional controllers can be connected to Daisy Chain via RS-485 communication for up to 60 axis control.

- **Providing a function to improve the precision of the stage**

When used along with the MMT Stage, it can calibrate the instrumental backlash, and can also be configured with a high-precision closed loop system through the 3D Encoder built into the MMT UVW Stage.

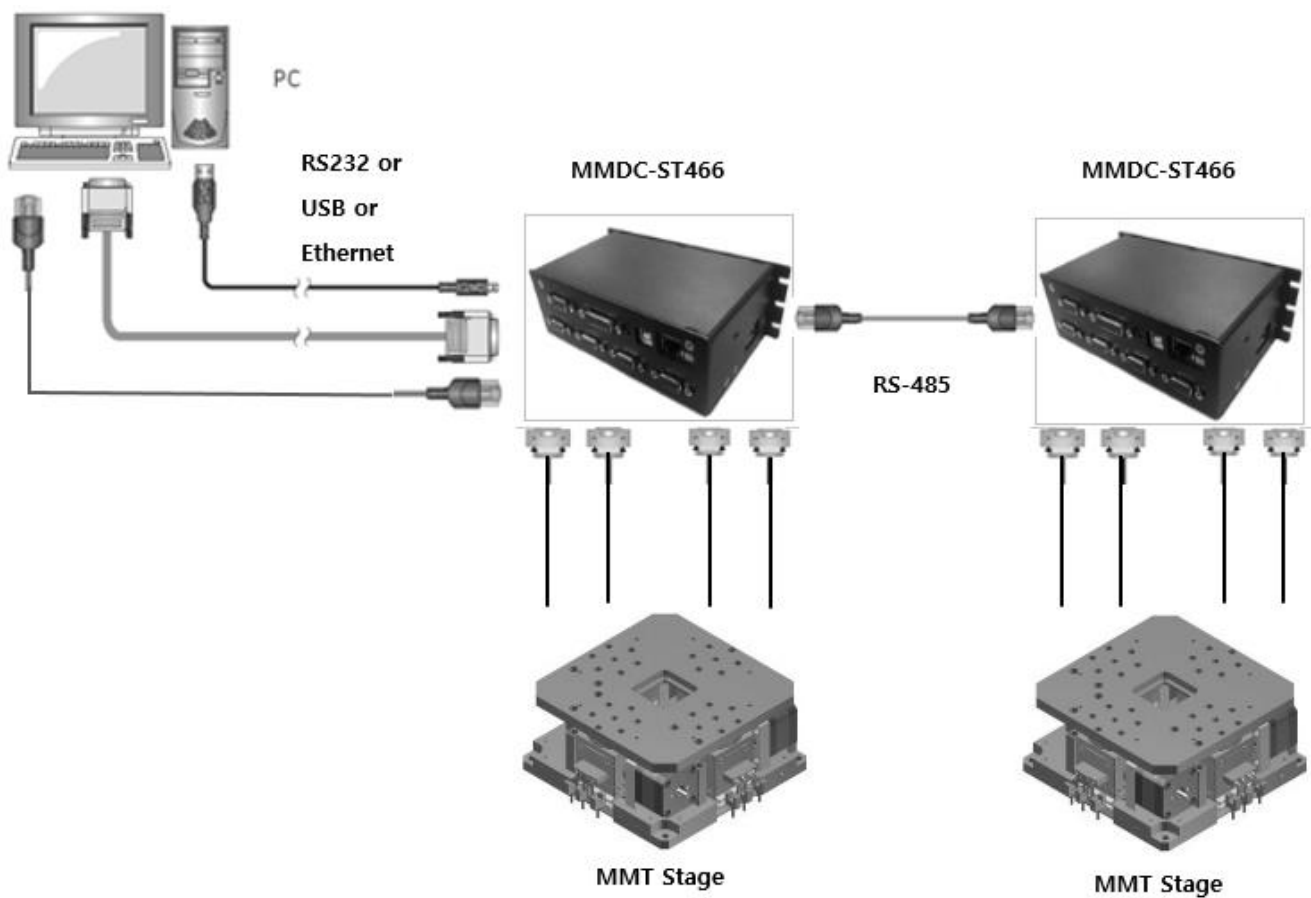
- **Superior space savings and cost efficiency**

Four drivers for step motor driving consist of an integral controller, which reduces wiring and saves space.

- MMDC-ST466-P is a product developed by adding DC24V power supplier to MMDC-ST466 products to improve space utilization
- MMMC-SM466 is attached to the MMT UVW Stage and do not require any space for complex wiring or controllers. Due to low space constraints, they are highly utilized.

System Configuration

Example)

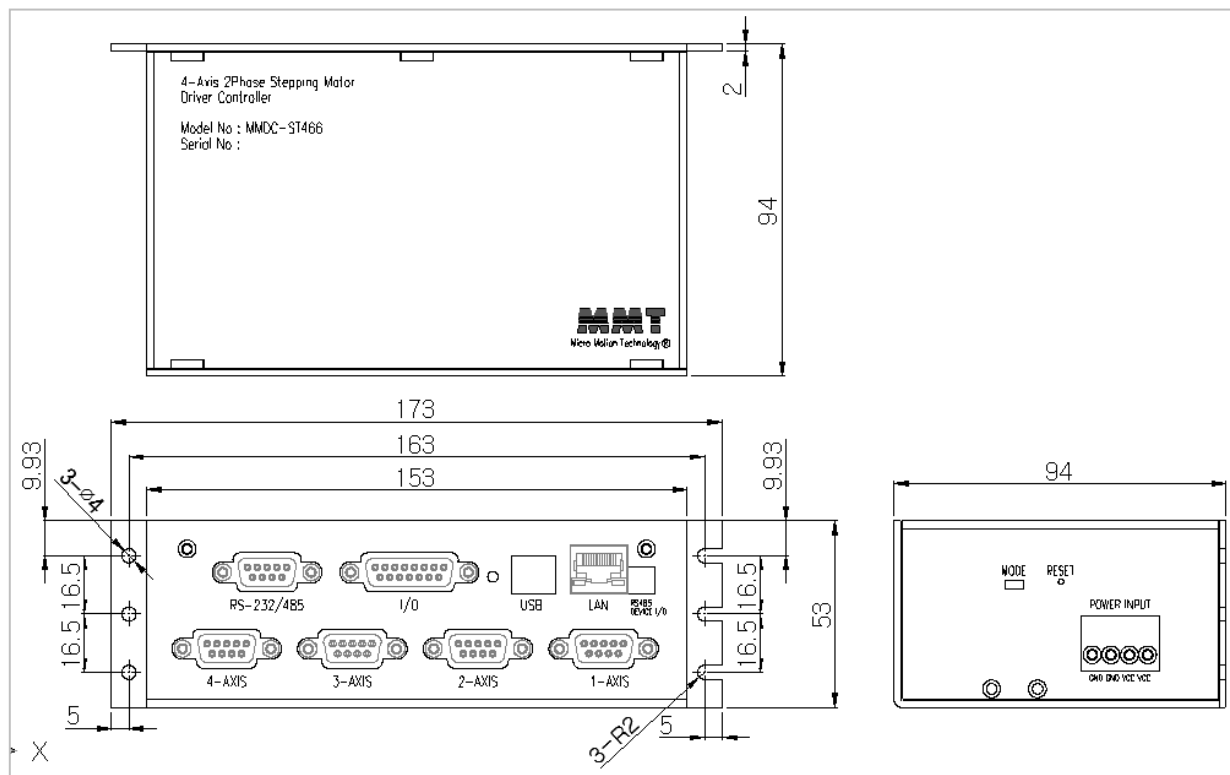


Specification

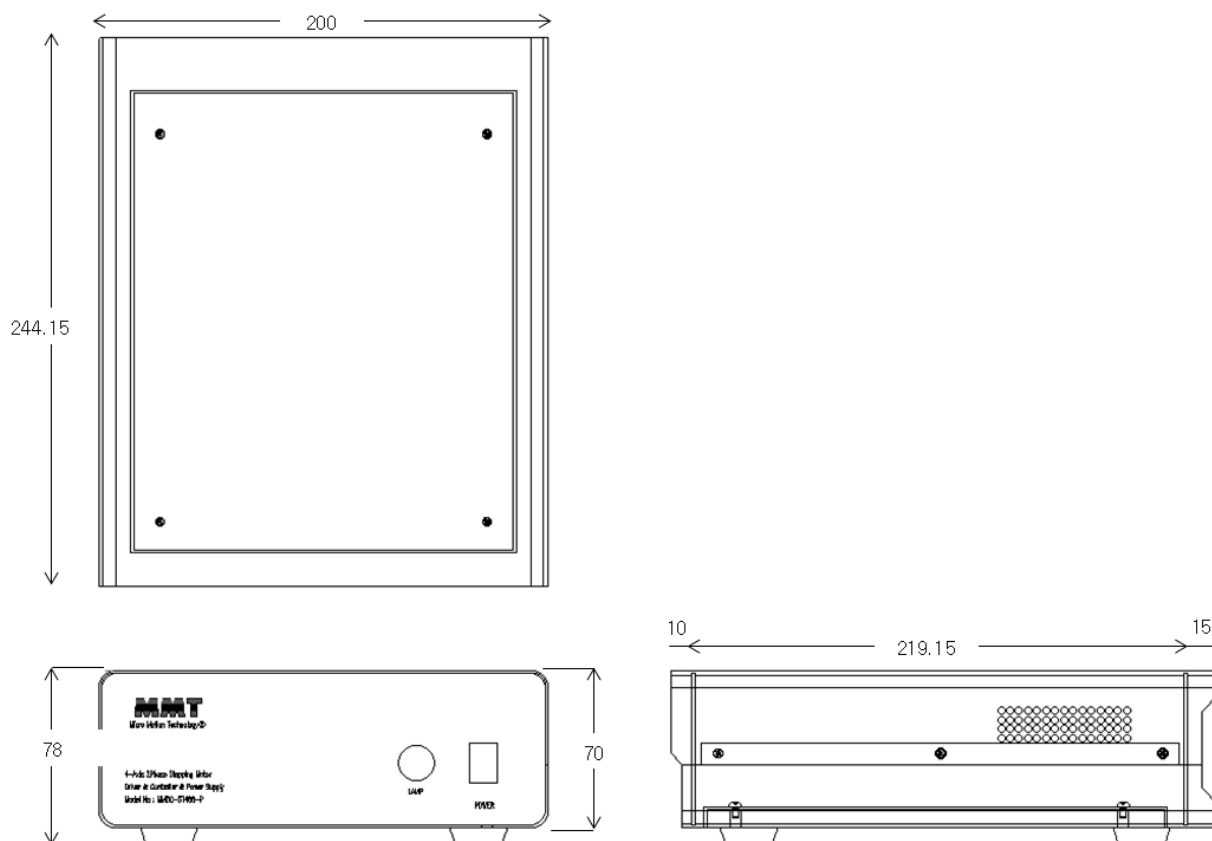
Product	MMDC-ST466	MMDC-ST466-P	MMDC-SM466
Type	Controller & Driver	Controller & Driver & Power Supplier	Stage Built-in (Built-in Type)
Dimension(mm)	W173 x D94 x H53	W200 x D244 x H70	W130 x D50 x H23 W70 x D50 x H18
Control Axis	4 (Maximum 60 Axis can be linked)		
Voltage	DC 24V	AC 85~264V	DC 24V
Watt	Below 120W		
Function	Absolute position, relative position drive, Simultaneous driving of 4 axes, Homing, repeat back and forth		
Microstep Resolution	51,200 microstep/Rev (※ Fixed Figure, Not-changeable)		
Stepper current	1.5A / Phase		
Sensor Signal	2 limits & 1 Home for each axis		2 limits for each axis (SMD Photo Microsensor)
Communication	RS232, Ethernet, USB2.0(B Type)		RS232, Ethernet
Control S/W	Provides basic S/W such as Jog, Limit, I/O, Programming, etc.		

Dimension

<MMDC-ST466>

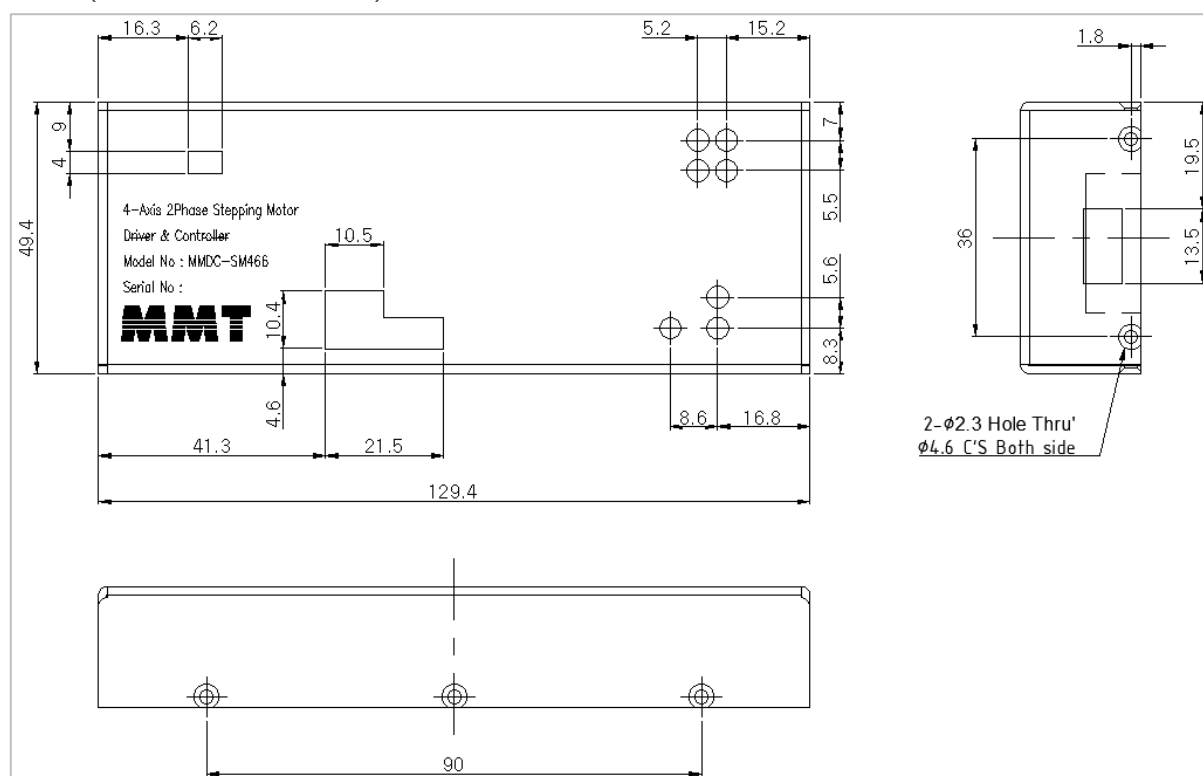


<MMDC-ST466-P>

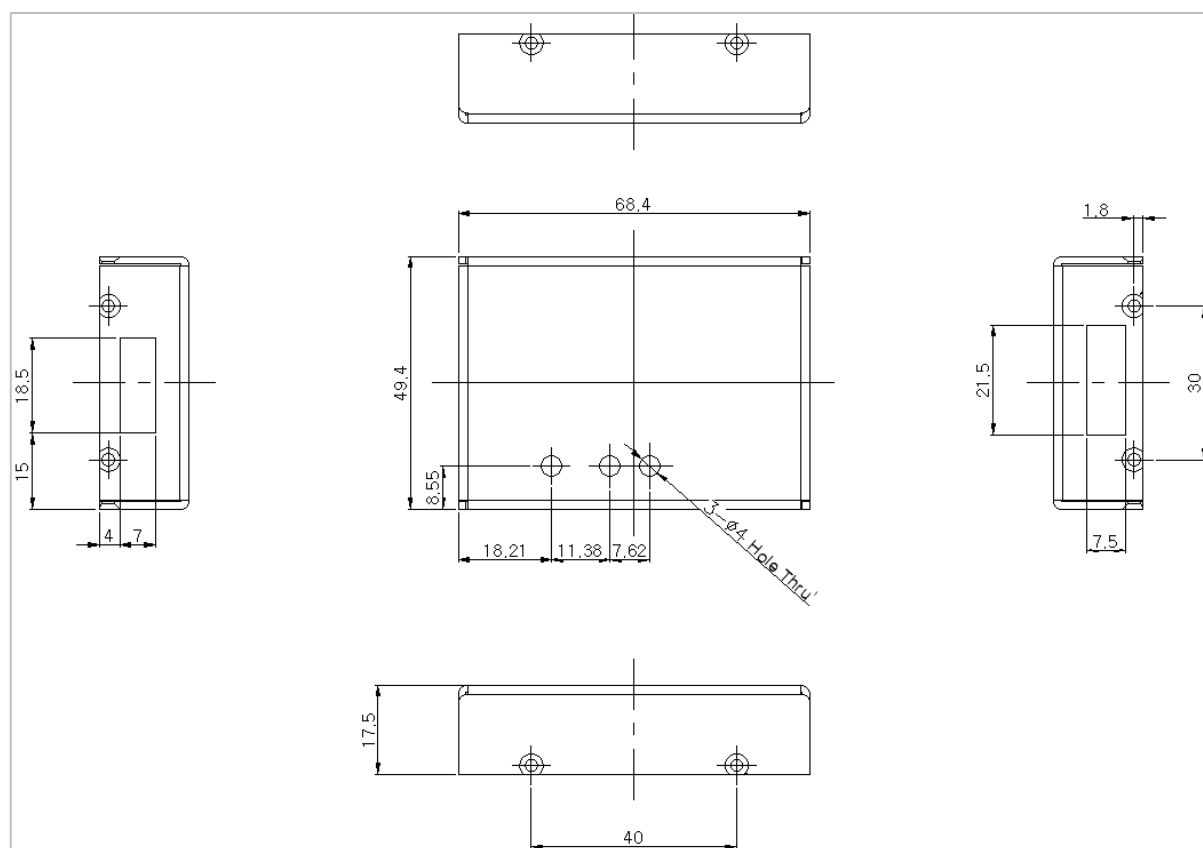


<MMDC-SM466>

1) Controller (Controller & Driver)



2) Driver



Part Name & Pin map

<MMDC-ST466>



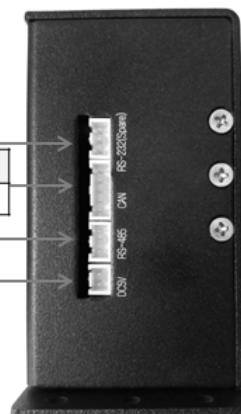
Name	Description
MODE	Mode switch - RUN : Run mode(Default) - PRG : Firmware Update Mode ※ It must be connected to RS-232 port in order to firmware update

Name	Description
RESET	Reset Button. ※ The reset button resets the state of the processor to its power-on condition

Name	Connector	Mating connector	Description	Pin	1	2	3	4
POWER INPUT	SLPH-300R-4PIN (SEOIL ELECTRONICS)	SLPS-300V-4PIN (SEOIL ELECTRONICS)	Supply Input (DC24V)	Function	GND	GND	24VIN	24VIN

Name	Connector	Mating connector	Description
RS-232 (Spare)	SMAW250-3 (YEONHO ELECTRONICS)	YH025-3 (YEONHO ELECTRONICS)	RS-232 Port
CAN	SMAW250-5	YH025-5	CAN PORT
RS-485	SMAW250-3	YH025-3	RS-485 Port
DC5V	SMAW250-2	YH025-2	DC5V Out

Pin	1	2	3
Function	GND	RXD	TXD
Pin	1	2	3
Function	GND	CANH	CANL
Pin	1	2	3
Function	GND	485 A	485 B
Pin	1	2	
Function	GND	5VOUT	



Name	Connector	Description	Pin	1	2	3	4	5	6	7	8	9
RS-232/485	DB9 Female	RS-232 및 RS-485 PORT	Function	N/A	TXD (RS232)	RXD (RS232)	N/A	GND (RS232)	485 A	485 B	485 TERMI A	485 TERMI B

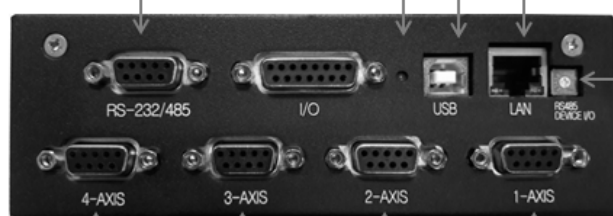
※ Caution! Do not connect in Pin 1,4,6,7,8,9 when use RS-232

Name	Description
LED	-Status monitor LED- Once Blink : Power on,HOST0:Not connected Twice Blink : Power on,HOST0:Connected Third Blink : Power on,HOST1:Not connected Forth Blink : Power on,HOST1:Connected

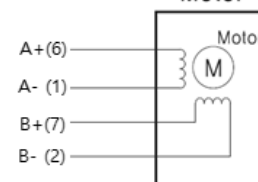
Name	Description
USB	USB Port

Name	Connector	Description
LAN	RJ-45	Ethernet Port

Name	Description
RS485 DEVICE I/D	RS485 Device (Module) ID Setting



Motor



Name	Connector	Description	Pin	1	2	3	4	5	6	7	8	9
1-AXIS	DB9 Female	1-Axis Motor and Limit/Home Sensor	Function	PHASE A-	PHASE B-	CW	HOME	CCW	PHASE A+	PHASE B+	GND	24VIN
2-AXIS	DB9 Female	2-Axis Motor and Limit/Home Sensor	(1 ~ 4axis Common use)	(Motor)	(Motor)	(Limit Input)	(Input)	(Limit Input)	(Motor)	(Motor)	(Limit/home power) Output	(Limit/home Power) Output
3-AXIS	DB9 Female	3-Axis Motor and Limit/Home Sensor										
4-AXIS	DB9 Female	4-Axis Motor and Limit/Home Sensor										

※ Allows you to use limit & home signal that is sinking(NPN)

Name	Connector	Description	Pin NO.	1	2	3	4	5	6	7	8
I/O	DB15 Female	Digital Input & Output	Function	OUT 6	OUT 4	OUT 2	IN 6	IN 4	IN 2	GND	24Vout
			In/Out	Output	Output	Output	Input	Input	Input	0V	Output
			Pin NO.	9	10	11	12	13	14	15	
			Function	OUT 5	OUT 3	OUT 1	IN 5	IN 3	IN 1	GND	
			In/Out	Output	Output	Output	Input	Input	Input	0V	
			Pin NO.								



<MMDC-ST466>

1) Controller (Controller & Driver)

<FRONT>

Name	Explanation
DEVICE ID	Dial Switch for device ID Setting while using RS485

Name	Description
RESET SW	Reset Button ※ Reset to the initial state before power input

Name	Connector	Relative Connector	Description
RS-232	YMW025-03R (Yeonho)	YMH025-03R (Yeonho)	FOR RS-232 Serial Communication

No.	1	2	3
Signal	GND	RXD	TXD

Name	Connector	Relative Connector	Description
PRG /DBG	YMW025-03R (Yeonho)	YMH025-03R (Yeonho)	FOR S/W Program Download

No.	1	2	3
Signal	GND	RXD	TXD

<Back>

Name	Relative Connector	Description
LAN	RJ-45	LAN Communication Port

Name	Connector	Relative Connector	Description
MOTOR	YMW025-04R	YMH025-04R	Motor Driving

Pin No.	1	2	3	4
Signal	AP(A+)	AN(A-)	BP(B+)	BN(B-)

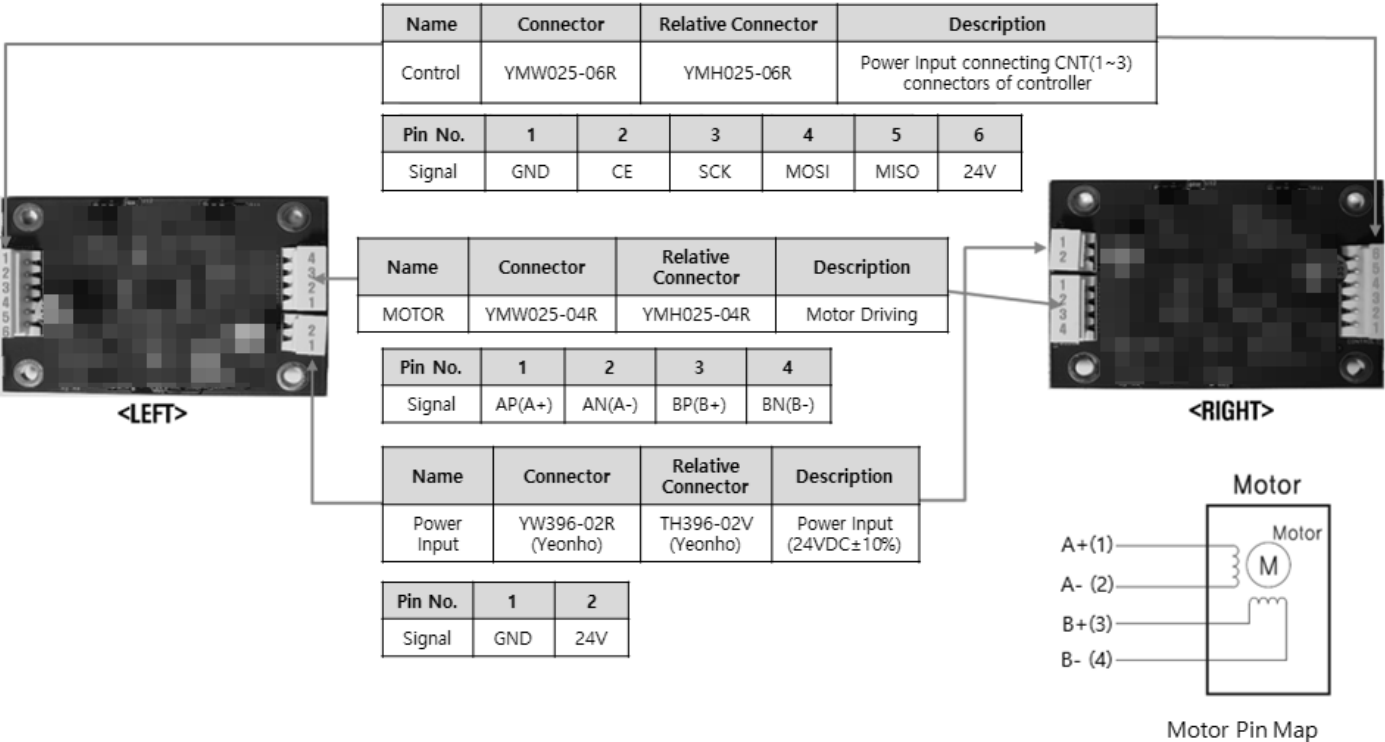
Name	Connector	Relative Connector	Description
CNT1	YMW025-06R	YMH025-06R	Power & Communication with 2 nd axis driver
CNT2	YMW025-06R	YMH025-06R	Power & Communication with 3rd axis driver
CNT3	YMW025-06R	YMH025-06R	Power & Communication with 4th axis driver

Name	Connector	Relative Connector	Description
Power Input	YW396-02R (Yeonho)	TH396-02V (Yeonho)	Power Input (24VDC±10%)

Pin No.	1	2
Signal	GND	24V

Pin No.	1	2	3	4	5	6
Signal	GND	CE	SCK	MOSI	MISO	24V

2) Driver



Network Setting

1. RS-232 Setting

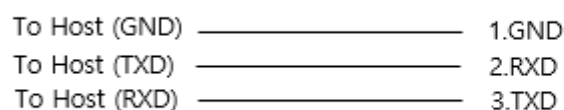
- ◆ Connect the RS-232 serial cable (customer preparation) to the COM PORT on the PC and the RS-232 PORT on the instrument, then set up COM PORT in the Device Manager.
- ◆ Set the COM PORT property of the PC to 115,200 bits per second, the data bit 8, parity, and flow control to None.
- ◆ If there is no RS-232 serial port on your PC, a separate converter (RS-232 To USB) is required

<MMDC-ST466>

※ **Caution!** Do not connect in Pin 1,4,6,7,8,9 when use RS-232 communication.



<MMDC-SM466>



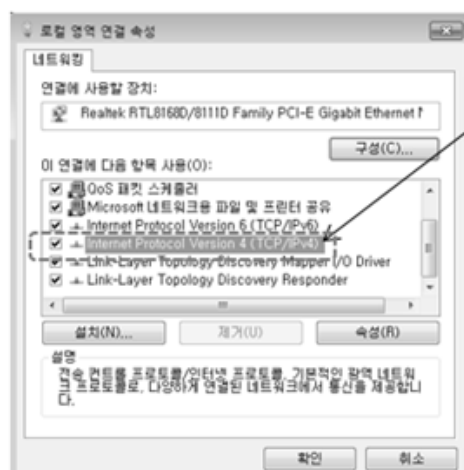
2. Ethernet Setting

- ◆ MMDC-ST466 supports industrial standard TCP/IP protocols. TCP/IP is a communication protocol.
- ◆ After you have prepared the LAN cable for Cross-Type, connect to the LAN PORT on your PC and the LAN PORT on the instrument, then set the IP address on the Local NETWORK setup screen on your PC.

(Controller IP set to Default : 192.168.0.123 / Port : 5001)

- ◆ Because the controller IP address is 192.168.0.123, the IP address of the local network of the PC must be assigned 192.168.0.###.

(### is any number from 0 to 255, except for 123. In addition, the Subnet Mask address on the local network of the PC must be assigned as 255.255.255.0)



◆ In the Local property on the Network Setup screen found at the Windows Control Panel, click "Internet Protocol Version 4"

◆ Set IP address and Subnet Mask address as shown in Example screen



3. RS-485 Setting

- ◆ Used when link each four-axis driver/controller each other, connectable up to 15 (60 axis) of the same controller.
- ◆ Connect the 485A, 485B signal line from the RS-232/485 port (DSUB9PS) or the RS-485 port to each instrument one by one parallel.
- ◆ If you do not have an RS-485 serial port on your PC, a separate converter (RS-485 to USB or RS-232) is required



Once connect resistance for the Controller & Driver connector's 8th (TERMI A) and 9th from outside, terminating resistance will be connected in innerside. If the products need to connect at the final step of communication cable connection, then set the terminating resistance.

※ **CAUTION!** The Baud rate setting of the drive modules connected within one segment must be set at 38,400 bps.

◆ Device (module) ID numbers must be different for additional controller connections.

Its range is from 0 to F (15) and Number 1 acts as the communication gateway for the HOST function.

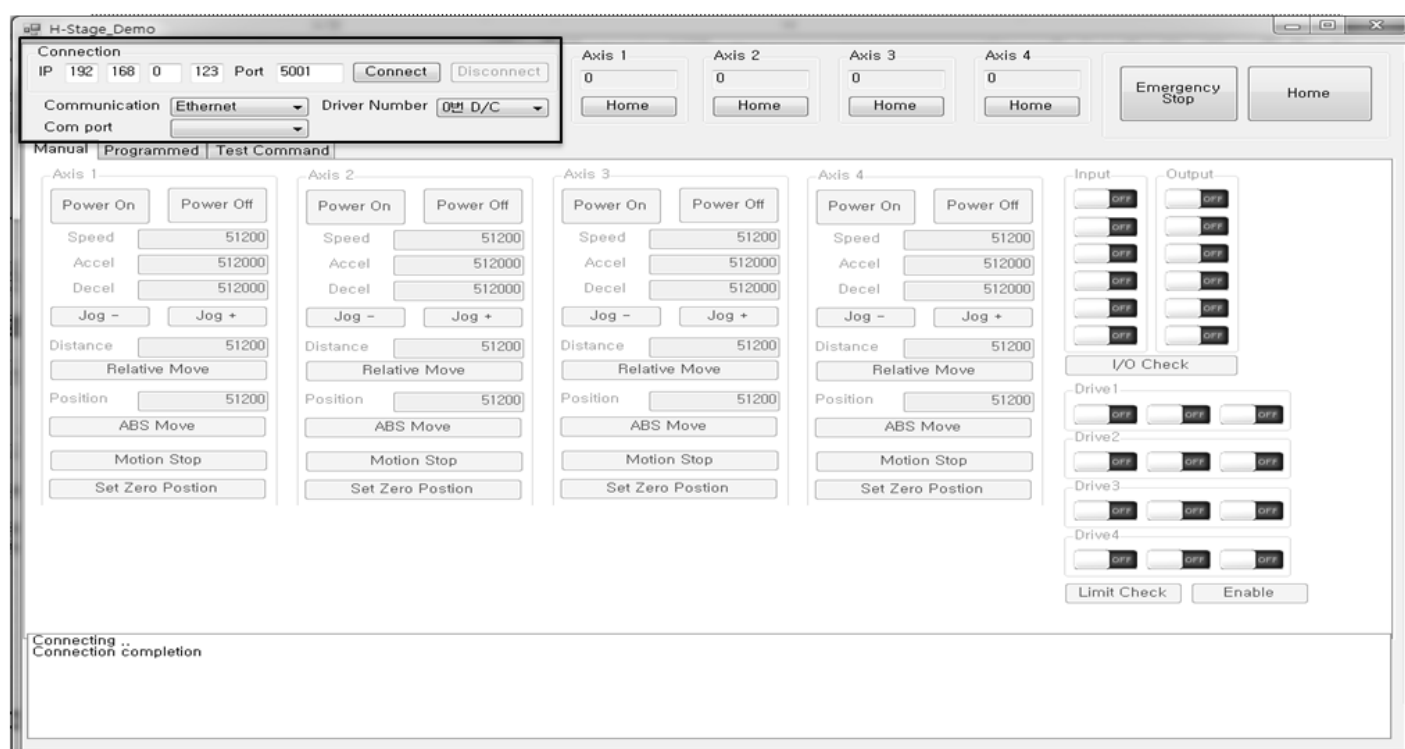


How to use the Control S/W

<The way of connection>

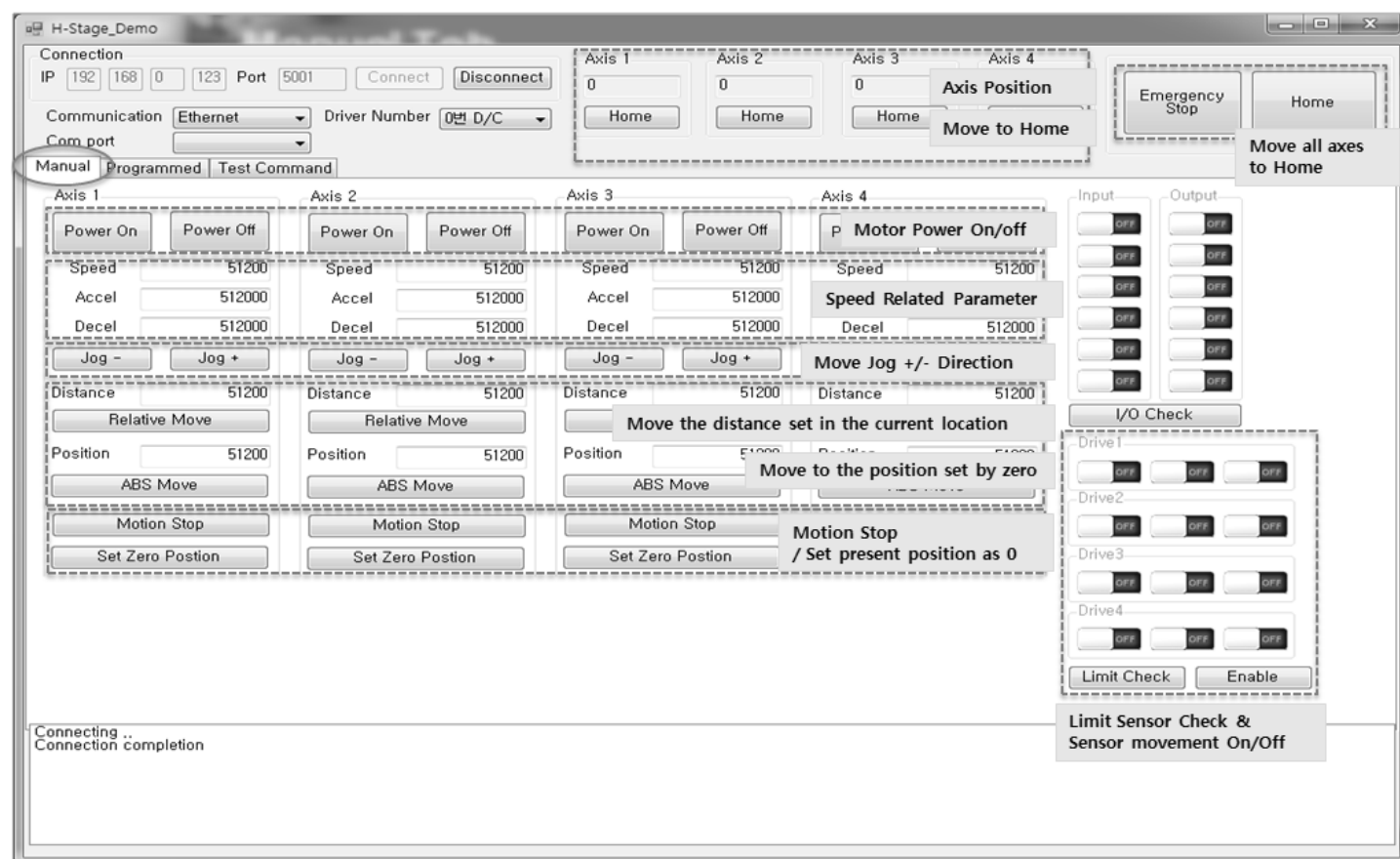
Connection

- IP & Port : Enter the IP and port number assigned to the controller to be accessed and click Connect (ex 192.168.0.123 / 5001)
- Communication : Choose Communication Method (Ethernet / Serial Port / USB (unsupported))
- Driver Number : Select the Device ID of the controller you want to connect to (Check the RS485 Device ID on the controller)
- Com port : Select a port connected while a serial communication connection



Manual Tab

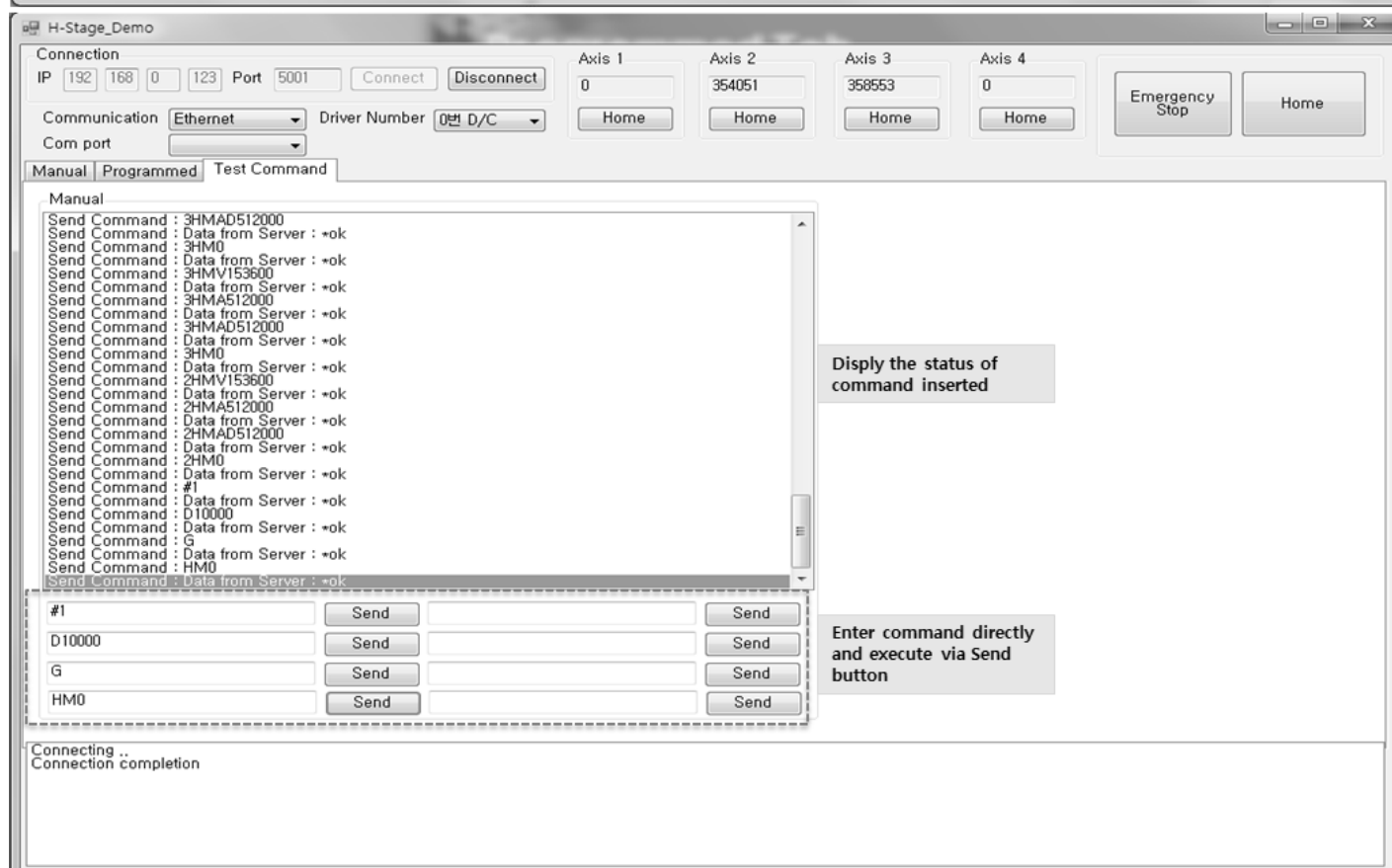
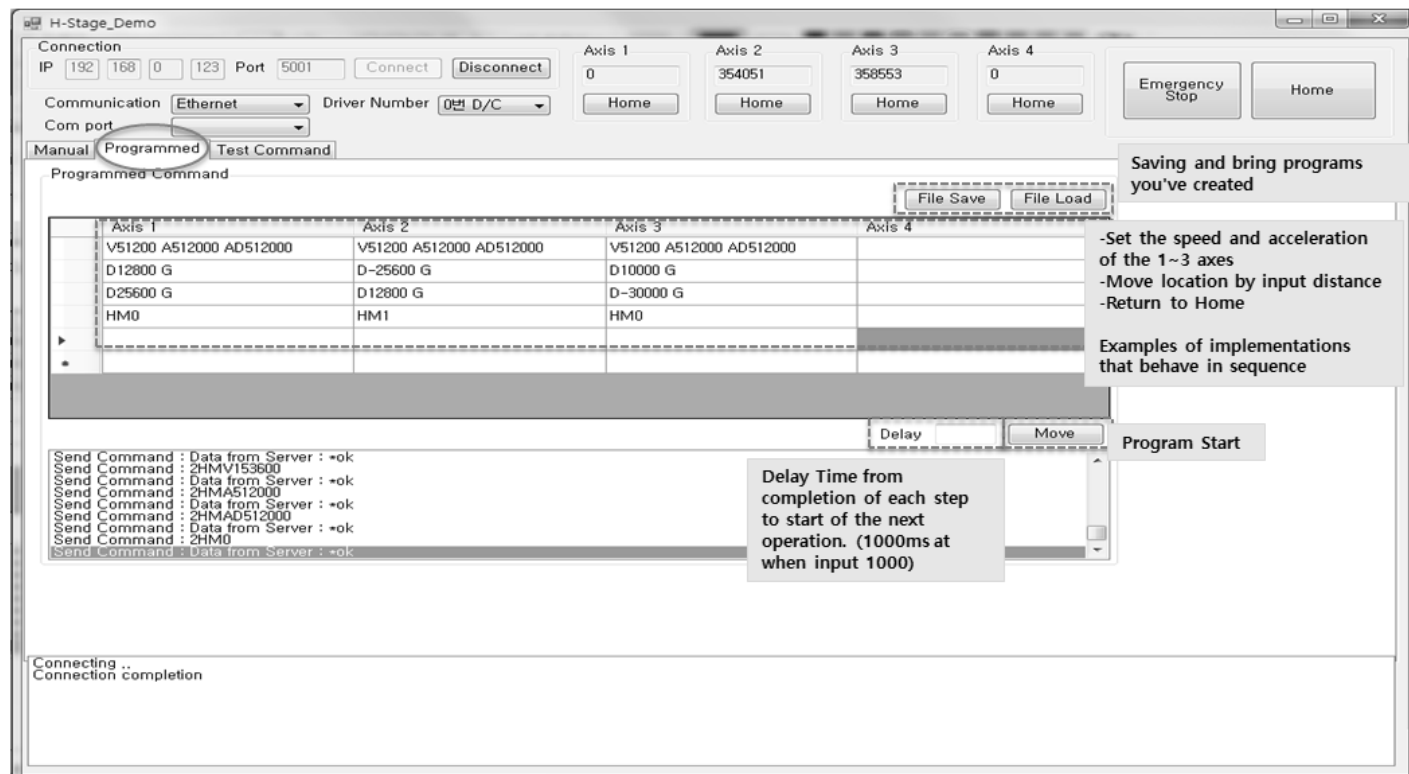
<Manual Tab User Guide>



Program Tab

<Programed Tab User Guide>

- Using the D/C command provided by MMT, it is simple to implement and operate as a program (Refer to the command list on the 11~29P of MMDC-ST466 Manual)



※ How to use control-related commands

The MMDC-ST466 provides commands such as motion-related and setup parameters. The command is sent to the ASCII code. The ASCII code consists of 1 to 4 easy-to-use characters, such as the Basic programming language, and is a command that can be executed immediately. Please be careful because it is divided into small letters and capital letters.

Commands can be used in a terminal program (ex Hyper Terminal etc.)

• Command

Motion					
G (GO)	Start motion	S (J/)	Stop motion	A (ACC)	Motor Acceleration
MA	Sets to Mode Absolute or Incremental Positioning	J	Move motor to Absolute Position.	AD (DEC)	Motor Deceleration
D (DIST)	Set the distance and direction of the move	V (VEL)	Motor Velocity	TEST	The round trip Motion
HMA (HOMACC)	Home Acceleration	HM (HOM)	Go Home	HMV (HOMVEL)	Home Velocity
HMA (HOMDEC)	Home Deceleration	HMVF (HOMVF)	Home Final Velocity	HML (HOMLVL)	Home Sensor Level
BLNB	Enable/Disable Backlash Compensation	HMO (HOMOFS)	Home Offset	HME (HOMEDGE)	Home Edge
BLDIR	Backlash direction	BLSTEP	Backlash steps	IP	Set the IP Address of the Controller
PORT	Set the Port Number of the Controller	net	Network Status		

Command Set

Configuration					
IH	Sets to Stop Current Of the Motor	IR	Sets to Run Current Of the Motor	REV (REVERSE)	Reverse the motor direction
PZ (POSZ)	Sets to Zero of Current Position	ST	Turns on/off the motor power	STATUS	Report a current status of driver
P (POS)	Report or Define the current position	IN	Input Status Report	HLSWAP	Hard limits Swap
HLD	Hard Limits Enable/Disable	SLD	Software Limits Enable/Disable	SLN	Negative Software limit Distance
SLP	Positive Software limit Distance	@	Send identical parameters to all axes connected to the communication line	&	Choice a Module (RS-485)
&!	Report All Connected modules(RS-485)	OUT	Immediate Output		

• Command Example

IP	IP Address		
Type	Setup	Valid	Read/Write
Syntax	IP 192,168,0,1	Units	None
Range	None	Default	192,168,0,123
Attributes	Buffered	See Also	PORT , net

This command sets the IP Address of the controller when using Ethernet Communication , Auto Saved

Examples:

IP 192,168,0,1 Sets the IP Address to 192,168,0,1
 IP *IP 192,168,0,1 Report the IP Address

※ **Caution!** Please change the IP address by connecting to RS-232 communication.

PORT	PORT Number		
Type	Setup	Valid	Read/Write
Syntax	PORT5001	Units	None
Range	None	Default	5001
Attributes	Buffered	See Also	IP , net

This command sets the PORT Number of the controller when using Ethernet Communication , Auto Saved

PORT5001 Sets the PORT Number to 192,168,0,1
 PORT *PORT5001 Report the PORT Number

※ **Caution!** Please change the IP address by connecting to RS-232 communication.

net	Network Status		
Type	Setup	Valid	Read
Syntax	net	Units	None
Range	None	Default	
Attributes	Buffered	See Also	IP , PORT

This command reports the status of Network

net * Reports the status of Network

[a]MA[p]	Sets to Mode Absolute or Incremental Positioning		
Type	Motion	Valid	Read/Write
Syntax	1MA0, 1MA1	Units	None
Range	0,1	Default	0
Attributes	Buffered	See Also	J

Sets to Mode Absolute or Incremental Positioning

Examples:

#4 Designate the current axis as #4
 MA1 Sets to Absolute mode of #4-Axis
 3MA1 Sets to Absolute mode of #3-Axis
 MA0 Sets to Incremental mode of #4-Axis
 3MA0 Sets to Incremental mode of #3-Axis
 M *4MA Report the Current mode

[a]V[p]	Motor Velocity		
Type	Motion	Valid	Read/Write
Syntax	1V500,V500,1V,V	Units	Steps/sec
Range	V=1~5000000	Default	512000
Attributes	Buffered	See Also	A,D,G

This command sets the slew speed of any axes.

Examples:

#4		Designate the current axis as #4
V10000		Sets velocity of #4-Axis to 10000steps/sec
1V51200		Sets velocity of #1-Axis to 51200steps/sec
3V51200		Sets velocity of #3-Axis to 51200steps/sec
3V	*3V51200	Reports the velocity rate of #3-Axis
V	*4V10000	Reports the velocity rate of current Axis

[a]A[p]	Motor Acceleration		
Type	Motion	Valid	Read/Write
Syntax	1A500,A500,1A,A	Units	Steps/sec^2
Range	p=10~50000000	Default	5120000
Attributes	Buffered	See Also	V,D,G

The Acceleration command sets the linear acceleration rate of the motors for independent moves

Examples:

#4		Designate the current axis as #4
A500000		Sets acceleration of #4-Axis to 500000steps/sec^2
1A50000		Sets acceleration of #1-Axis to 50000steps/sec^2
3A15000		Sets acceleration of #3-Axis to 15000steps/sec^2
3A	*3A15000	Reports the acceleration rate of #3-Axis
A	*4A500000	Reports the acceleration rate of current Axis

[a]AD[p]	Motor Deceleration		
Type	Motion	Valid	Read/Write
Syntax	1AD500,AD500,1AD,A D	Units	Steps/sec^2
Range	p=10~50000000	Default	5120000
Attributes	Buffered	See Also	A,V,D,G

The Deceleration command sets the linear deceleration rate of the motors for independent moves

Examples:

#4		Designate the current axis as #4
AD500000		Sets deceleration of #4-Axis to 500000steps/sec^2
1AD50000		Sets deceleration of #1-Axis to 50000steps/sec^2
3AD15000		Sets deceleration of #3-Axis to 15000steps/sec^2
3AD	*3AD15000	Reports the deceleration rate of #3-Axis
AD	*4AD500000	Reports the deceleration rate of current Axis

[a]D[p]	Set the distance and direction of the move		
Type	Motion	Valid	Read/Write
Syntax	1D500,D300,D	Units	Steps
Range	p=-200000000 ~ 200000000	Default	51200
Attributes	Buffered	See Also	A,AD,V,MA,G

This command sets the distance and direction of the next move

#4		Designate the current axis as #4
MA1		Sets to absolute mode of #4-Axis
D-5000		Sets distance to 5,000 steps & negative direction
D5000		Sets distance to 5,000 steps & positive direction
A50000		Sets acceleration of #4-Axis to 50000steps/sec^2
AD50000		Sets deceleration of #4-Axis to 50000steps/sec^2
G		Execute the move
		After motion stop
P	*5000	Report the current position
MA0		Sets to incremental mode of #4-Axis
G		Execute the move
		After motion stop
P	*10000	Report the current position
D	*4D5000	Report the current axis number & distance

[a]G	Go, Execute the move		
Type	Motion	Valid	Write
Syntax	1G,G	Units	None
Range	None	Default	None
Attributes	Buffered	See Also	S

The G command initiates motion based on previous motion parameters. G will move the motor the number of steps specified with D command

Examples:

#4		Designate the current axis as #4
MA1		Sets to absolute mode of 4 Axis
D5000		Sets distance to 5,000 steps & positive direction
A50000		Sets acceleration of #4-Axis to 50,000step/sec^2
AD50000		Sets deceleration of #4-Axis to 50,000step/sec^2
G		Executes the move (Go)
		After motion stop
P	*5000	Report the current position
MA0		Sets to incremental mode of #4-Axis
G		Execute the move
		After motion stop
P	*10000	Report the current position

[a]S	Stop		
Type	Motion	Valid	Write
Syntax	1S,S	Units	None
Range	None	Default	None
Attributes	Buffered	See Also	G

This command decelerates motion to a stop using the last defined deceleration value.

Examples:

#4		Designate the current axis as #4
J+		Absolute Move to positive direction of #4-Axis
S		Stop the motion of #4-Axis

[a]J[p]	Absolute Move		
Type	Motion	Valid	Read/Write
Syntax	J=10000,J+=500	Units	None
Range	None	Default	None
Attributes	Buffered	See Also	MA,D,G

Move motor to Absolute Position.

Examples:

#4	Designate the current axis as #4
J=10000	Move to 10,000 steps of positive direction
J+=500	Move to 500 steps of positive direction from the current position
J-=500	Move to 500 steps of negative direction from the current position
J/	Stop motion
J+	Absolute Move to positive direction of #4-Axis
J-	Absolute Move to negative direction of #4-Axis

[a]P[p]	Report or Define the current Position		
Type	Motion	Valid	Read/Write
Syntax	1P,1P5000,P1000	Units	Steps
Range	-200000000 ~ 200000000	Default	0
Attributes	Immediate	See Also	PZ

This command reports the current position or defines the current position.

Examples:

#4	Designate the current axis as #4
MA1	Sets to Absolute mode of #4-Axis
D5000	Sets distance to 5,000 steps & positive direction
G	Executes the move
P	Report the current position
P1230	Define the current position to 1230 steps
P	Report the current position

[a]PZ[p]	Position ZERO		
Type	Motion	Valid	Read/Write
Syntax	PZ	Units	Steps
Range	None	Default	0
Attributes	Immediate	See Also	P

Define the current position to Zero

Examples:

#4	Designate the current axis as #4
MA1	Sets to Absolute mode of #4-Axis
D5000	Sets distance to 5,000 steps & positive direction
G	Executes the move
P	Report the current position
PZ	Define the current position to Zero
P	Report the current position

[a]HLD[p]	Hardware Limit Enable or Disable		
Type	Setup	Valid	Read/Write
Syntax	1HLD3,2HLD1,1HLD	Units	None
Range	0,1,2,3,none	Default	0
Attributes	Buffered	See Also	SLD

HLD command allows you to enable/disable the end-of-travel limit switch protection.

If you want motion without wiring the limits, use HLD3

Examples:

#4	Designate the current axis as #4
HLD0	Enable CW & CCW limits
HLD1	Disable CW limits
HLD2	Disable CCW limits
HLD3	Disable CW and CCW limits
HLD *4HLD3	Report the current status

⌘ **Caution!** It is recommended that the limits sensor be connected with Normal closed and the home sensor with Normal open.

[a]HLSWAP[p]	H/W Limit SWAP		
Type	Setup	Valid	Read/Write
Syntax	HLSWAP0	Units	None
Range	0,1	Default	0
Attributes	Buffered	See Also	

Inverting the CW and CCW of the hardware limits

Examples:

#4	Designate the current axis as #4
HLSWAP0	Not Inverting the CW and CCW of the hardware limits
HLSWAP1	Inverting the CW and CCW of the hardware limits
HLSWAP *4HLSWAP1	Report the current status

[a]SLD[p]	S/W Limit Enable or Disable		
Type	Setup	Valid	Read/Write
Syntax	SLD1	Units	None
Range	0,1,2,3,none	Default	0
Attributes	Buffered	See Also	SLP,SLN

SLD command allows you to enable/disable the software limit switch protection

Examples:

#4	Designate the current axis as #4
SLD0	Enable CW & CCW S/W limits
SLD1	Disable CW S/W limits
SLD2	Disable CCW S/W Limits
SLD3	Disable CW and CCW S/W limits
SLD *4SLD3	Report the current status

[a]SLP[p]	Positive S/W Limit Position		
Type	Setup	Valid	Read/Write
Syntax	SLP10000	Units	None
Range	0~10000000	Default	0
Attributes	Buffered	See Also	SLD.SLN

Attributes	Parameter	Default	Unit
Sets to the position of the software limit of positive direction	POS_LIMIT	0	mm

Examples:

#4	Designate the current axis as #4
SLP 50000	The S/W limit is active at the 50000 PULSE position in the positive direction
SLP 10000	The S/W limit is active at the 10000 PULSE position in the positive direction
SLP *4SLP10000	Report the current status

[a]SLN[p]	Negative S/W Limit Position		
Type	Setup	Valid	Read/Write
Syntax	SLN50000	Units	None
Range	0~10000000	Default	0
Attributes	Buffered	See Also	SLD.SLP

Sets to the position of the software limit of negative direction

Examples:

#4	Designate the current axis as #4
SLN50000	The S/W limit is active at the 50000 PULSE position in the negative direction
SLN10000	The S/W limit is active at the 10000 PULSE position in the negative direction
SLN*4SLN10000	Report the current status

IN	Input		
Type	I/O	Valid	Read
Syntax	IN	Units	Bit
Range	0,1	Default	000000
Attributes	Immediate	See Also	OUT

IN 6

↓

IN 011001

IN 1

↓

ON : 1
OFF : 0

This command reports the status of the 6 inputs

Examples:

IN	*IN011001	Status of all the Inputs are reported #1,4,5 input status is ON #2,3,6 input status is OFF
----	-----------	--

OUT[p6]..[p1]	Immediate Output			
Type	I/O	Valid	Read/Write	
Syntax	OUT1xx011	Units	Bit	
Range	1,0,x	Default	000000	
Attributes	Immediate	See Also	IN	

OUT6

↓

OUT010xx1

↓

OUT1

↓

OUT10xx1

ON : 1
OFF : 0

This command immediately sets the 6 output bits

Examples:

OUTxxxxx1	Turns on output 1 , leaves other outputs unaffected
OUTxxxx01	Turns on output 1 , Turns off output 2 ,leaves other outputs unaffected
OUT101010	Turns off output 1,3,5, Turns on output 2,4,6
OUT1xxxxxx	Turns on output 6 , leaves other outputs unaffected

[a]REV[p]	Reverse the moto rdirection		
Type	Setup	Valid	Read
Syntax	1R0,1R1,	Units	None
Range	0,1	Default	None
Attributes	Buffered	See Also	

This command defines the direction of the motor

Examples:

```
#2          Designate the current axis as #2
REV0       Sets move to CW
REV1       Sets move to CCW
R          Report The Status
           *2R0001_0010_0000_0100_1001
           bit16 -> 1          Changed the motor direction
```

[a]HM[p]	Go Home with specified direction		
Type	Motion	Valid	Read/Write
Syntax	1HM1,HM0	Units	
Range	1,0	Default	0
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

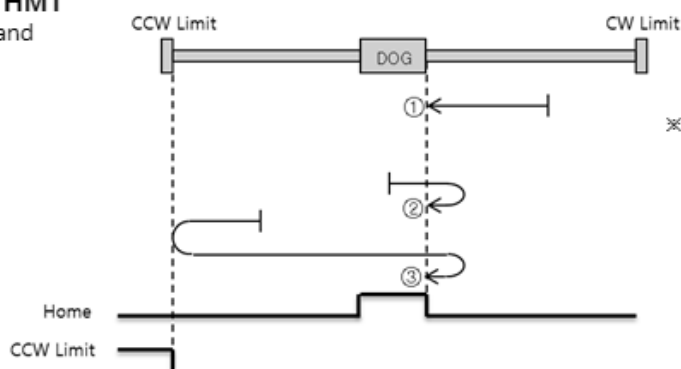
HM command search for the home limit switch in the positive or negative direction

Examples:

```
#2
HMA500000  Sets go home acceleration to 500000 steps/sec^2
HMAD500000 Sets go home deceleration to 500000 steps/sec^2
HMA500000  Sets go home velocity to 50000 steps/sec
HMVF500    Sets go home final velocity to 500 steps/sec
HMO15000   After the home position is reached, sets current position to 15,000 steps
HM1        This command will move the motor to Search for home to CCW direction
HM0        This command will move the motor to Search for home to CW direction
```

✂ It is recommended that the limits sensor be connected with Normal closed and the home sensor with Normal open.

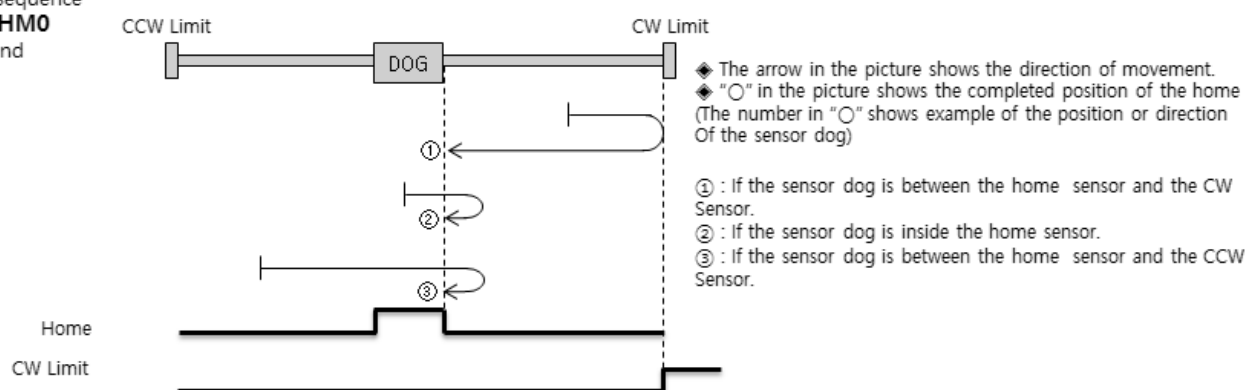
Home Sequence
Of the **HM1**
Command



✂ The arrow in the picture shows the direction of movement.
"O" in the picture shows the completed position of the home
(The number in "O" shows example of the position or direction
Of the sensor dog)

- ① If the sensor dog is between the home sensor and the CW Sensor.
- ② If the sensor dog is inside the home sensor.
- ③ If the sensor dog is between the home sensor and the CCW Sensor.

Home Sequence
Of the **HMO**
Command



[a]HMA[p]	Set to Home Acceleration		
Type	Motion	Valid	Read/Write
Syntax	1HMA5000,HMA	Units	Steps/sec ²
Range	10~500000	Default	None
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

[a]HMAD[p]	Set to Home Deceleration		
Type	Motion	Valid	Read/Write
Syntax	1HMAD5000,HMAD	Units	Steps/sec ²
Range	10~500000	Default	None
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

[a]HMA[p]	Set to Home Velocity		
Type	Motion	Valid	Read/Write
Syntax	1HMA5000,HMA	Units	Steps/sec
Range	10~500000	Default	None
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

[a]HMO[p]	Home Offset		
Type	Motion	Valid	Read/Write
Syntax	1HMO1000,HMO	Units	Steps
Range	-200000000 ~ 200000000	Default	0
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

After the home position is reached, sets current position to you want

Examples:

#2	Designate the current axis as #2
HMA500000	Sets go home acceleration to 500000 steps/sec^2
HMAD500000	Sets go home deceleration to 500000 steps/sec^2
HMV50000	Sets go home velocity to 50000 steps/sec
HMVF500	Sets go home final velocity to 500 steps/sec
HMO15000	After the home position is reached, sets current position to 15,000 steps
HM1	This command will move the motor to Search for home to Positive direction

[a]HML[p]	Reverse Home sensor Level		
Type	Setup	Valid	Read/Write
Syntax	1HMO,HML1	Units	0,1
Range	0,1,None	Default	0
Attributes	Buffered	See Also	HMA,HMAD,HMV,HMVF,HMO,HM1

Reverse Home sensor Level (Normal closed <-> Normal open). Follow the sensor wiring and Default status is recommended.

Examples:

#2	Designate the current axis as #2
HML1	Reverses the home sensor
HMAD500000	Sets go home deceleration to 500000 steps/sec^2
HMV50000	Sets go home velocity to 50000 steps/sec
HMVF500	Sets go home final velocity to 500 steps/sec
HMO15000	After the home position is reached, sets current position to 15,000 steps
HM1	This command will move the motor to Search for home to Positive direction

[a]BLENB[p]	Enable/Disable Backlash Compensation		
Type	Setup	Valid	Read/Write
Syntax	1BLENB1,1BLENB0	Units	None
Range	0,1	Default	0
Attributes	Buffered	See Also	BLDIR,BLSTEP,D,G

This command enables compensation of any mechanical backlash

Examples:

#1	Designate the current axis as #1
BLSEP500	500 steps are added to motion when changing with direction of BLS
BLDIR0	Sets negative direction of backlash
BLENB1	Enables backlash compensation
D10000	Sets distance to 10,000 steps & positive direction
G	Execute the move (motor travels 10,500 steps and travels 500 steps to opposite direction)
BLENB0	Disables backlash compensation

[a]BLDIR[p]	Backlash direction		
Type	Setup	Valid	Read/Write
Syntax	1BLDIR1,1BLDIR0	Units	None
Range	0,1	Default	0
Attributes	Buffered	See Also	BLENB,BLSTEP,D,G

This command sets direction of the backlash

Examples:

#1
 BLSTEP500 Designate the current axis as #1
 500 steps are added to motion when changing with direction of BLS
 BLDIR1 Sets positive direction of backlash
 BLENB1 Enables backlash compensation

[a]BLSTEP[p]	Backlash steps		
Type	Setup	Valid	Read/Write
Syntax	1BLSTEP500	Units	None
Range	p=1~10000	Default	0
Attributes	Buffered	See Also	BLENB,BLDIR,D,G

This command sets steps of backlash

Examples:

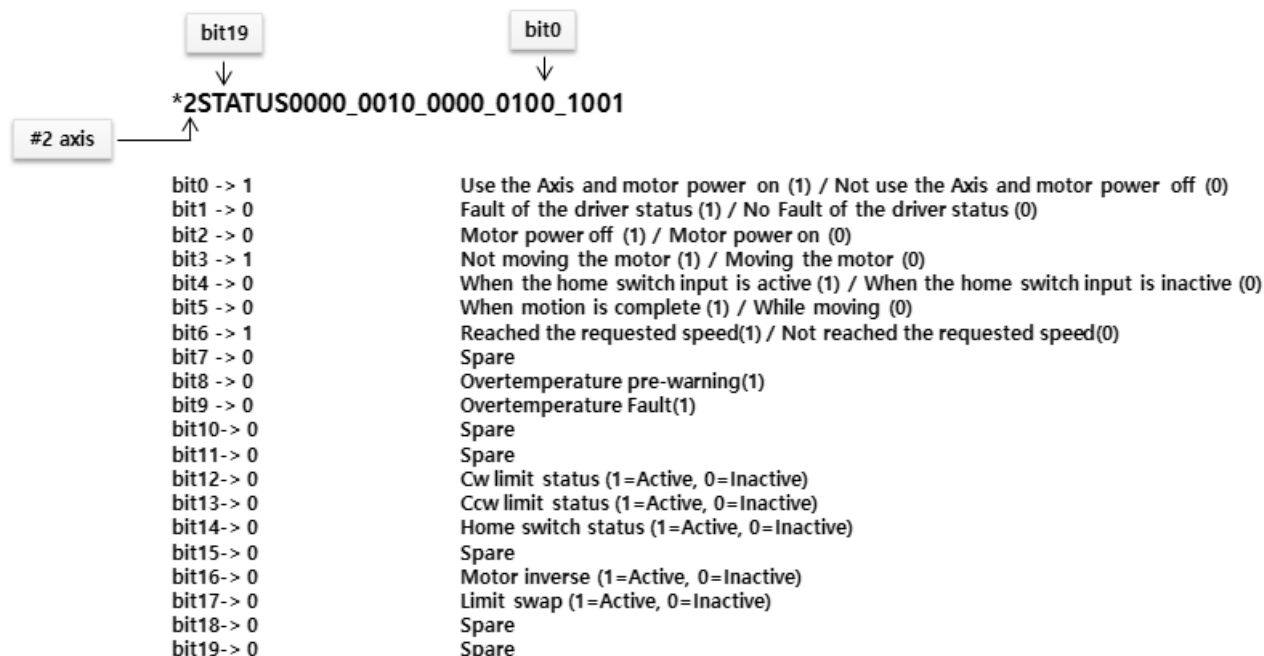
#1
 BLSTEP500 Designate the current axis as #1
 500 steps are added to motion when changing with direction of BLS
 BLDIR0 Sets negative direction of backlash
 BLENB1 Enables backlash compensation

[a]STATUS	Report status of driver		
Type	Status	Valid	Read
Syntax	1STATUS,2STATUS, STATUS	Units	Bit
Range	None	Default	None
Attributes	Immediate	See Also	

This command reports the status of the driver

Examples:

#2
 STATUS Designate the current axis as #2
 Reports the status of the current axis



Multi Module (RS-485)

&[M]	Select the module on which to perform the command.(RS 485)		
Type	select	Valid	
Syntax	&1, &12	Units	Non
Range	none	Default	none
Attributes	Immediate	See Also	&!, @

Select the modules to be operated / Start with module 0

&3		Select the module number 3
	*&3ok	Response if module number 3 is connected
		No response if there is no connection
&15		Select the module 15
	*&15ok	Response if module number 15 is connected
		No response if there is no connection

&!	All modules currently connected to the communications line are displayed. (RS 485)		
Type	SET UP	Valid	
Syntax	&!	Units	Non
Range	none	Default	none
Attributes	Immediate	See Also	&!, @

All modules currently connected to the communications line are displayed

&!		
	*&3#4	Module number 3 is connected
	*&12#4	Module number 12 is connected
	*&15#4	Module number 15 is connected

@command	Send the same parameters to all axes of the module where the communication is active. (RS 485)		
Type	command	Valid	Read/Write
Syntax	@A10000,@V500	Units	Non
Range	none	Default	none
Attributes	Immediate	See Also	A,AD,V,D,....

Send the same parameters to all axes of the module where the communication is active

&3		Select the module 3
@A5000		Sets acceleration of All Axes to 5000steps/sec^2
@A	*&3A5000,5000,5000,5000	Reports the acceleration rate of All Axes
@V5000		Sets velocity of All Axes to 5000steps/sec

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